

This paper covers Differential Geometry, Barrett O'Neill,  
with the sub Analysis on Manifolds, Munkres.

## Chapter 1. Calculus on Euclidean Space.

Def. A real-valued  $f: \mathbb{R}^3 \rightarrow \mathbb{R}$  is called smooth if:

$$\frac{\partial^k f}{\partial x_1^{k_1} \partial x_2^{k_2} \partial x_3^{k_3}} \text{ exists and continuous for all } k_1, k_2, k_3 \geq 0.$$

And, given two smooth functions  $f$  and  $g$ , define

$$(f+g)(p) = f(p) + g(p) \quad \text{and} \quad (fg)(p) = f(p) \cdot g(p).$$

Def. A tuple  $V_p = (p, v) \in \mathbb{R}^3 \times \mathbb{R}^3$  is called a tangent vector,

consisting of the point part  $p$  and vector part  $v$ .

And, the set  $T_p(\mathbb{R}^3) \stackrel{\text{def}}{=} \{(p, v) \mid v \in \mathbb{R}^3\}$  is called the tangent space of  $\mathbb{R}^3$  at  $p$ .

Def. A mapping  $V: \mathbb{R}^3 \rightarrow \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3) : p \mapsto V_p$  is called vector field on  $\mathbb{R}^3$ .

Given two vector field  $V$  and  $W$ , and real-valued  $f$  on  $\mathbb{R}^3$ , define.

$$(V+W)(p) = V(p) + W(p) \quad \text{and} \quad (fV)(p) = f(p)V(p).$$

In particular, the vector fields on  $\mathbb{R}^3$  define by

$$U_1(p) = (1, 0, 0)_p \quad \text{and} \quad U_2(p) = (0, 1, 0)_p \quad \text{and} \quad U_3(p) = (0, 0, 1)_p \quad \text{for each } p \in \mathbb{R}^3$$

are called the natural frame field on  $\mathbb{R}^3$ .

Lemma. Given a vector field  $V$  on  $\mathbb{R}^3$ , there are unique real-valued  $V_1, V_2, V_3$  on  $\mathbb{R}^3$

$$\text{s.t.} \quad V = V_1 U_1 + V_2 U_2 + V_3 U_3.$$

Particularly, the functions  $V_1, V_2, V_3$  are called the Euclidean coordinate functions of  $V$ .

Proof. By definition, for each point  $p \in \mathbb{R}^3$ , the vector part of  $V(p)$  depends on  $p$ , so, denote the vector part of  $V(p)$  as  $(V_1(p), V_2(p), V_3(p))$ . Now,

$$V(p) = (V_1(p), V_2(p), V_3(p))_p = V_1(p) U_1(p) + V_2(p) U_2(p) + V_3(p) U_3(p).$$

And, the uniqueness is given by the comparing the entries of tuples.

**Def.** Let  $f$  be a smooth real-valued function on  $\mathbb{R}^3$ , and  $V_p$  be a tangent vector. The limit  $V_p[f] \stackrel{\text{def}}{=} \left. \frac{d}{dt}(f(p + tV)) \right|_{t=0} \left( = \lim_{t \rightarrow 0} \frac{f(p + tV) - f(p)}{t} \right)$

is called the directional derivative of  $f$  with respect to  $V_p$ .

**Lemma.** Given a tangent vector  $V_p = (V_1, V_2, V_3)_p$  on  $\mathbb{R}^3$  and smooth function  $f$ ,

$$V_p[f] = \sum_{i=1}^3 V_i \cdot \frac{\partial f}{\partial x_i}(p)$$

**Proof.** Let  $P = (p_1, p_2, p_3) \in \mathbb{R}^3$  be given. Then

$$p + tV = (p_1 + tV_1, p_2 + tV_2, p_3 + tV_3)$$

So that  $f(p + tV) = f(p_1 + tV_1, p_2 + tV_2, p_3 + tV_3)$

Now, by the chain rule and the fact that  $\frac{d}{dt}(p_i + tV_i) = V_i$ ,

$$V_p[f] = \left. \frac{d}{dt}(f(p + tV)) \right|_{t=0} = \sum_{i=1}^3 \frac{\partial f}{\partial x_i}(p) V_i. \quad \square$$

**Note:** In other words,

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} \cdot \frac{dx_1}{dt} + \frac{\partial f}{\partial x_2} \cdot \frac{dx_2}{dt} + \frac{\partial f}{\partial x_3} \cdot \frac{dx_3}{dt}$$

**Thm.** Let  $f$  and  $g$  be functions on  $\mathbb{R}^3$ ,  $V_p$  and  $W_p$  be tangent vectors, and  $a, b \in \mathbb{R}$ .

$$1) (aV_p + bW_p)[f] = aV_p[f] + bW_p[f]$$

$$2) V_p[af + bg] = aV_p[f] + bV_p[g]$$

$$3) V_p[f g] = V_p[f] \cdot g(p) + f(p) \cdot V_p[g].$$

**Proof.** These are obtained by the Lemma above.

**Corollary.** Given two vector fields  $V$  and  $W$  on  $\mathbb{R}^3$  and real-valued  $f, g, h$  on  $\mathbb{R}^3$ ,

$$1) (fV + gW)[h] = fV[h] + gW[h]$$

$$2) V[af + bg] = aV[f] + bV[g] \quad \text{for all } a, b \in \mathbb{R}.$$

$$3) V[f g] = V[f] \cdot g + f V[g].$$

Proof.

$$(av_p + bw_p)[f] = \sum_{i=1}^{\text{Lemma 3}} (av_i + bw_i) \cdot \frac{\partial f}{\partial x_i}(p) = a \sum v_i \frac{\partial f}{\partial x_i}(p) + b \sum w_i \frac{\partial f}{\partial x_i}(p).$$

Sec 1.4 Curves in  $\mathbb{R}^3$ .  $[I]$  is any open interval.

Def. A differentiable function  $\alpha: I \rightarrow \mathbb{R}^3$  is called a Curve.

Moreover, for  $\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t))$ , the tangent vector

$$\alpha'(t) = \left( \frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{\alpha(t)}$$

is called the velocity vector of  $\alpha$  at  $t$ .

Examples for Curves and the velocity vector.

## The parametrization

Def. Let  $\alpha: I \rightarrow \mathbb{R}^3$  be a curve defined on an open interval  $I$ .

If  $h: J \rightarrow I$  is a differentiable function on an open interval  $J$ , then the composite  $\beta = \alpha \circ h: J \rightarrow \mathbb{R}^3$  is a curve called a reparametrization of  $\alpha$  by  $h$ .

Lemma. If  $\beta$  is the reparametrization of  $\alpha$  by  $h$ , then the velocity is

$$\beta'(s) = \frac{dh}{ds}(s) \cdot \alpha'(h(s))$$

Proof. Denote  $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ . Then, by the definition,

$$\beta(s) = (\alpha \circ h)(s) = (\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s)))$$

Now, by the definition of velocity and the chain rule,

$$\beta'(s) = (\alpha_1'(h(s)) \cdot h'(s), \alpha_2'(h(s)) \cdot h'(s), \alpha_3'(h(s)) \cdot h'(s))$$

$$= h'(s) \cdot (\alpha_1'(h(s)), \alpha_2'(h(s)), \alpha_3'(h(s)))$$

$$= h'(s) \cdot \alpha'(h(s)).$$

Lemma. Let  $\alpha$  be a curve in  $\mathbb{R}^3$  and let  $f$  be a differentiable function on  $\mathbb{R}^3$ .

Then  $\alpha'(t)[f] = \frac{d(f \circ \alpha)}{dt}(t)$

Proof.

$\Pi$ -form.

Def. A map  $\varphi: \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3) \rightarrow \mathbb{R}$  is called a 1-form on  $\mathbb{R}^3$  if;

$$\varphi(av_p + bw_p) = a \cdot \varphi(v_p) + b \cdot \varphi(w_p) \text{ for all } a, b \in \mathbb{R} \text{ and } v_p, w_p \in T_p(\mathbb{R}^3), \text{ for each } p.$$

And, for a fixed  $p \in \mathbb{R}^3$ , the restriction  $\varphi_p: T_p(\mathbb{R}^3) \rightarrow \mathbb{R}: v_p \mapsto \varphi(v_p)$  is a linear functional.

Moreover, given two 1-forms  $\varphi_1, \varphi_2$  and  $f, g: \mathbb{R}^3 \rightarrow \mathbb{R}$ , define

$$(\varphi_1 + \varphi_2)(v_p) \stackrel{\text{def}}{=} \varphi_1(v_p) + \varphi_2(v_p) \text{ and } (f\varphi)(v_p) = f(p) \cdot \varphi(v_p), \text{ for each } v_p \in \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3)$$

And given two vector fields  $V, W$ ,

$$\varphi(fV + gW) \stackrel{\text{def}}{=} f\varphi(V) + g\varphi(W)$$

$$\text{and } (f\varphi_1 + g\varphi_2)(V) \stackrel{\text{def}}{=} f\varphi_1(V) + g\varphi_2(V).$$

Def. If  $f: \mathbb{R}^3 \rightarrow \mathbb{R}$  is differentiable, the 1-form

$$df: \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3) \rightarrow \mathbb{R}: v_p \mapsto v_p[f]$$

is called the differential of  $f$ .

Example. Consider the projections  $\pi_i: \mathbb{R}^3 \rightarrow \mathbb{R}: (x_i) \mapsto x_i$  ( $i=1,2,3$ ). Then,

$$d\pi_i(v_p) = v_p[\pi_i] = \sum_{j=1}^3 v_j \cdot \frac{\partial \pi_i}{\partial x_j}(p) = v_i.$$

Lemma. Let  $\varphi$  be a 1-form on  $\mathbb{R}^3$ . Then,

$$\varphi = \sum f_i dx_i \text{ where } f_i = \varphi(v_i)$$

Proof. For fixed  $p \in \mathbb{R}^3$ ,  $\varphi_p = \sum \varphi_p \circ v_p \cdot (dx_i)_p$ .

Note: for fixed  $p \in \mathbb{R}^3$ , the tangent space  $T_p(\mathbb{R}^3)$  is isomorphic to  $\mathbb{R}^3$ .

So, the dual sp.  $(T_p(\mathbb{R}^3))^*$  is a set of 1-forms at  $a$ . Moreover,

the triple  $\{(dx_1)_p, (dx_2)_p, (dx_3)_p\}$  is a dual basis for  $(T_p(\mathbb{R}^3))^*$ , corresponding to  $\{v_1(p), v_2(p), v_3(p)\}$ .

## Chapter 2. Frame field.

Def. Given tangent vectors  $V_p, W_p$  at  $p \in \mathbb{R}^3$ , a tangent vector

$$V_p \times W_p \stackrel{\text{def}}{=} \begin{vmatrix} U_1(p) & U_2(p) & U_3(p) \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} \stackrel{(*)}{=} \begin{aligned} &U_1(p)(V_2 W_3 - V_3 W_2) \\ &- U_2(p)(V_1 W_3 - V_3 W_1) \\ &+ U_3(p)(V_1 W_2 - V_2 W_1). \end{aligned}$$

is called the cross product of  $V_p$  and  $W_p$ .

Lemma. Given  $V_p$  and  $W_p$  in  $T_p(\mathbb{R}^3)$ ,  $V_p \times W_p$  is orthogonal to both  $V_p$  and  $W_p$ ,

and 
$$\|V_p \times W_p\|^2 = (V_p \cdot V_p) \cdot (W_p \cdot W_p) - (V_p \cdot W_p)^2.$$

In other say,  $\|V \times W\| = \|V\| \cdot \|W\| \cdot \sin \theta$

Proof. Since  $U_1, U_2, U_3$  are orthonormal,

$$V \cdot (V \times W) = \begin{vmatrix} V_1 & V_2 & V_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0 \quad \text{and} \quad W \cdot (V \times W) = \begin{vmatrix} W_1 & W_2 & W_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0.$$

Therefore, the orthogonality is proved. And, to show the second assertion,

$$(V \cdot V) \cdot (W \cdot W) - (V \cdot W)^2 = (V_1^2 + V_2^2 + V_3^2)(W_1^2 + W_2^2 + W_3^2) - (V_1 W_1 + V_2 W_2 + V_3 W_3)^2$$

$$= \sum_{1 \leq i, j \leq 3} V_i^2 \cdot W_j^2 - \left( \sum_{1 \leq i \leq j \leq 3} V_i W_i V_j W_j \right)$$

$$= \sum_{1 \leq i, j \leq 3} V_i^2 \cdot W_j^2 - \left( \sum_{i=1}^3 V_i^2 W_i^2 + 2 \cdot \sum_{1 \leq i < j \leq 3} V_i W_i V_j W_j \right)$$

$$= \sum_{1 \leq i < j \leq 3} V_i^2 \cdot W_j^2 - 2 \cdot \sum_{1 \leq i < j \leq 3} V_i W_i V_j W_j$$

Meanwhile,

$$\|V \times W\|^2 = (V_2 W_3 - V_3 W_2)^2 + (V_1 W_3 - V_3 W_1)^2 + (V_1 W_2 - V_2 W_1)^2$$

$$\begin{aligned} &= V_2^2 W_3^2 + V_3^2 W_2^2 - 2 V_2 W_2 V_3 W_3 \\ &+ V_1^2 W_3^2 + V_3^2 W_1^2 - 2 V_1 W_1 V_3 W_3 \\ &+ V_1^2 W_2^2 + V_2^2 W_1^2 - 2 V_1 W_1 V_2 W_2 \end{aligned}$$

These are same.  $\square$

## Sec 2.2 Curves.

Recall: a differentiable function  $\alpha: I \rightarrow \mathbb{R}^3$  is called a Curve on an open interval  $I$ .

Def. Given a Curve  $\alpha: I \rightarrow \mathbb{R}^3$ ,

$$\alpha'(t) \stackrel{\text{def}}{=} \left( \frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{d\alpha}$$

is called the velocity of  $\alpha$ , and  $\|\alpha'(t)\|$  is called the speed of  $\alpha$ .

And, for  $[a, b] \subset I$ ,  $\int_a^b \|\alpha'(t)\| dt$

is called the arc length of  $\alpha$  from  $t=a$  to  $t=b$ .

Def. A curve  $\alpha: I \rightarrow \mathbb{R}^3$  is called regular if;

$$\alpha'(t) \neq 0 \text{ for all } t \in I.$$

Thm. Given a regular Curve  $\alpha$  in  $\mathbb{R}^3$ ,

there exists a reparametrization  $\beta$  of  $\alpha$  such that  $\|\beta'(s)\| = 1$  for all  $s$ .

Proof. Let  $\alpha: I \rightarrow \mathbb{R}^3$  be a regular Curve. Let  $a \in I$  be fixed.

Define  $S(t) = \int_a^t \|\alpha'(u)\| du$ . Since  $\|\alpha'\| \neq 0$ , the  $S(t)$  is monotone increasing on  $I$ .

Therefore, there is an inverse function  $t: J \rightarrow I$  s.t.  $S(t(x)) = x$  for all  $x \in J$ ,

where  $J = S[I]$ . Now, for  $\beta = \alpha \circ t$ ,

$$\beta'(s) = \frac{dt}{ds}(s) \cdot \alpha'(t(s)) \text{ and } \|\beta'(s)\| = \underbrace{\frac{dt}{ds}(s)}_{>0} \|\alpha'(t(s))\| = \frac{dt}{ds}(s) \cdot \frac{ds}{dt}(t(s)) = 1. \quad \square$$

Def. Given a curve  $\alpha: I \rightarrow \mathbb{R}^3$ , the function

$$\gamma: I \rightarrow \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3); t \mapsto V_{\alpha(t)}$$

is called a vector field on Curve  $\alpha$ .

## Sec 2.3 Frenet Formula.

Def. Let  $\beta: I \rightarrow \mathbb{R}^3$  be a unit-speed curve.

A vector field  $\beta'$  is called the unit tangent vector field on  $\beta$ , denote  $T = \beta'$ .

The derivative  $T' = \beta''$  is called the Curvature vector field of  $\beta$ .

The value  $\kappa(s) = \|T'(s)\|$  ( $s \in I$ ) is called the Curvature function of  $\beta$ .

To restriction s.t.  $\kappa > 0$ ,  
the unit-vector field  $N = \frac{T'}{\kappa}$  is called the principal normal vector field of  $\beta$ .

The cross product  $B = T \times N$  on  $\beta$  is called the binormal vector field of  $\beta$ .

Lemma. Let  $\beta$  be a unit-speed curve in  $\mathbb{R}^3$  with  $\kappa > 0$ ,

Then, the three vector fields  $T, N$ , and  $B$  on  $\beta$  are unit vector fields that are mutually orthogonal at each point.

Proof. By definition,  $\|T\| = \|\beta'\| = 1$ . And since the curvature  $\kappa > 0$ ,  $\|N\| = \frac{1}{\kappa} \cdot \|T'\| = 1$ .

And, since  $\|T\|^2 = T \cdot T$  is constant,  $2T \cdot T' = 0$ , so  $T \perp N$ .

therefore the binormal  $\|B\| = \|T \times N\| = \|T\|^2 \cdot \|N\|^2 - (T \cdot N)^2 = 1 - 0 = 1$ .

In summary,  $T = \beta'$ ,  $N = T'/\kappa$ ,  $B = T \times N$ , and  $\{T, N, B\}$  is orthonormal.

Since  $B \cdot T = 0$ ,  $B' \cdot T + B \cdot T' = 0$  by differentiate. Therefore,

$$B' \cdot T = -B \cdot T' = -B \cdot \kappa N = 0$$

And since  $B$  is unit,  $B' \cdot B = 0$ , thus, there is a scalar value function  $\tau$  s.t.

$$B' = -\tau N$$

, called the torsion. Now

Thm. Let  $\beta$  be a unit-speed curve in  $\mathbb{R}^3$  with curvature  $\kappa > 0$  and torsion  $\tau$ ,

$$\begin{aligned} T' &= \kappa N \\ N' &= -\kappa T + \tau B \\ B' &= -\tau N \end{aligned} \quad + \quad \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof.

Curve.

Def. A differentiable function  $\alpha: (a, b) \rightarrow \mathbb{R}^3$  is called a Curve.

And the tangent vector

$$\alpha'(t) \stackrel{\text{def}}{=} \left( \frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right) \stackrel{\text{def}}{=} \sum_{i=1}^3 \frac{d\alpha_i}{dt}(t) \cdot e_i(dt).$$

is called the velocity vector of  $\alpha$  at  $t$ .

If  $h: (c, d) \rightarrow (a, b)$  is a differentiable, then the composition

$$\beta = \alpha \circ h: J \rightarrow \mathbb{R}^3$$

is called a reparametrization of  $\alpha$  by  $h$ .

Lemmu. If  $\beta$  is the reparametrization of  $\alpha$  by  $h$ , then

$$\beta'(s) = h'(s) \cdot \alpha'(h(s)).$$



### Chapter 3. Euclidean Geometry.

Preliminary: Mappings.

Def. Given a function  $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ , define  $f_i = \pi_i \circ F$  for each  $1 \leq i \leq m$ . Then

$$F(p) = (f_1(p), \dots, f_m(p)) \text{ for all } p \in \mathbb{R}^n$$

these  $f_1, \dots, f_m$  are called the Euclidean coordinate functions of  $F$ .

If the coordinate functions are differentiable, then  $F$  is a mapping from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ .

Def. Given a Curve  $\alpha: I \rightarrow \mathbb{R}^n$  and mapping  $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ , a Composition

$$\beta: F \circ \alpha: I \rightarrow \mathbb{R}^m$$

is called the image of  $\alpha$  under  $F$ .

Def. Let  $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$  be a mapping define

$$F_*: \bigcup_{p \in \mathbb{R}^n} T_p \mathbb{R}^n \rightarrow \bigcup_{p \in \mathbb{R}^n} T_{F(p)} \mathbb{R}^m: V_p \mapsto \beta'(0)$$

where  $\beta: (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^m: t \mapsto F(p + tV)$ .

Prop. Let  $F = (f_1, \dots, f_m)$  be a mapping from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ .

Given a tangent vector  $V_p \in T_p(\mathbb{R}^n)$ ,

$$F_*(V_p) = (V_p[f_1], \dots, V_p[f_m])_{F(p)}$$

Proof. Since  $\beta(t) = (f_1(p + tV), \dots, f_m(p + tV))$ ,

$$\left. \frac{d}{dt} f_i(p + tV) \right|_{t=0} = V_p[f_i], \text{ so prove directly by the definition. } \square$$

Note: for fixed  $p \in \mathbb{R}^n$ ,  $F_{*p}: T_p(\mathbb{R}^n) \rightarrow T_{F(p)}(\mathbb{R}^m)$  is called the tangent map of  $F$  at  $p$ .  
Moreover, the  $F_{*p}$  is linear map.

Sec 3.2. The tangent map of an Isometry.

Thm. Let  $F$  be an isometry of  $\mathbb{R}^3$  with decomposition  $F = TC$ . Then

$$F_*(V_p) = (C \cdot V)_{F(p)}$$

# Chapter 4. Calculus on a surface.

Def. Let  $D \subseteq \mathbb{R}^2$  be an open set.

A one-to-one regular mapping  $X: D \rightarrow \mathbb{R}^3$  is called a Coordinate Patch.  
 $X_{*p}$  is 1-1 for each  $p$ .

A subset  $M \subseteq \mathbb{R}^3$  is called a Surface if:

for each  $p \in M$ , there exists a patch  $X$  such that  $N \subseteq \text{Im } X$ ,

where  $N$  is a neighborhood of  $p$  in  $M$ .

Example. Let  $S^2 = \{(p_1, p_2, p_3) \in \mathbb{R}^3 \mid \sqrt{p_1^2 + p_2^2 + p_3^2} = 1\}$  be given.

Find a proper patch in  $S^2$  covering a neighborhood of the North Pole  $(0, 0, 1)$ .

Let  $D = \{(u, v) \mid u^2 + v^2 < 1\}$ , and define

$$X: D \rightarrow \mathbb{R}^3: (u, v) \mapsto (u, v, \underbrace{\sqrt{1-u^2-v^2}}_f)$$

To show this  $X$  satisfies the Patch Condition, calculate the Jacobian

$$J(X) = \begin{pmatrix} \frac{\partial u}{\partial u} & \frac{\partial u}{\partial v} \\ \frac{\partial v}{\partial u} & \frac{\partial v}{\partial v} \\ \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{u}{f} & -\frac{v}{f} \end{pmatrix}$$

Trivially  $\text{rank } J = 2$ . So, by the inverse function theorem,  $X_{*p}$  is locally one-to-one.

And,  $X^{-1}: X[D] \rightarrow D: (p_1, p_2, p_3) \mapsto (p_1, p_2)$  is an inverse of  $X$ , so one-to-one.

In special case: Given differentiable  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $M = \{(x, y, f(x, y)) \mid x, y \in \mathbb{R}^2\}$  is a surface.

Via the Monge patch  $X(u, v) = (u, v, f(u, v))$ .

Patch Computation.

Def. Given a patch  $X: D \rightarrow \mathbb{R}^3$ , for fixed  $(u_0, v_0) \in D$ , define

$$u \mapsto X(u, v_0) \quad \text{and} \quad v \mapsto X(u_0, v)$$

are called  $u, v$  - Parameter Curves, respectively.

And, partial velocities of  $X$  at  $(u_0, v_0)$  as

$$X_u(u_0, v_0) \stackrel{\text{def}}{=} \left( \frac{\partial X}{\partial u}(u_0, v_0) \right)_{(u_0, v_0)} \quad \text{and} \quad X_v(u_0, v_0) \stackrel{\text{def}}{=} \left( \frac{\partial X}{\partial v}(u_0, v_0) \right)_{(u_0, v_0)}$$

Def. A regular mapping  $X: D \rightarrow \mathbb{R}^3$  s.t.  $X[D] \subseteq M$  for the surface  $M$  is called a parametrization of the region  $X[D]$  in  $M$ .

Thm. Given a mapping  $X: D \rightarrow \mathbb{R}^3$  where  $D \subseteq \mathbb{R}^2$  is open,

$X$  is regular if and only if the cross product  $X_u \times X_v \neq 0$  for all  $p \in D$ .

Proof. By definition,  $X$  is regular if and only if given  $p \in D$ , the Jacobian

$DX(p)$  is one-to-one, that is,  $\ker DX(p) = \{0\}$ , or  $\text{rank } DX(p) = 2$ .

The geographical patch in the sphere.

$$\text{Let } \Sigma \stackrel{\text{def}}{=} \left\{ (x, y, z) \in \mathbb{R}^3 \mid \sqrt{x^2 + y^2 + z^2} = r \right\} \text{ where } r > 0.$$

Define a patch

$$X: (-\pi, \pi) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow (u, v) \mapsto (r \cdot \cos u \cdot \cos v, r \cdot \sin u \cdot \cos v, r \cdot \sin v)$$

$$= r \cdot \cos v \cdot (\cos u, \sin u, 0) + r \cdot \sin v \cdot (0, 0, 1)$$

$$= r \cdot \cos v \cdot \cos u \cdot e_1 + r \cdot \cos v \cdot \sin u \cdot e_2 + r \cdot \sin v \cdot e_3.$$

Calculate the partial velocities

$$X_u(u, v) = (-r \cdot \sin u \cdot \cos v, r \cdot \cos u \cdot \cos v, 0)$$

$$X_v(u, v) = (-r \cdot \cos u \cdot \sin v, -r \cdot \sin u \cdot \sin v, r \cdot \cos v).$$

Surface Criterion.

Let  $g: \mathbb{R}^3 \rightarrow \mathbb{R}$  be differentiable, and  $c \in \mathbb{R}$ , and let  $M = \{(x, y, z) \in \mathbb{R}^3 \mid g(x, y, z) = c\}$ .  
If the differential  $dg$  is not zero for all  $p \in M$ , then  $M$  is a surface.